Employee Engagement & Burnout Analysis Dashboard

List of Contents

- Introduction
- Executive Summary
- Background & Bussiness Context
- Objectives
- Dataset Overview
- Data Preparation & Feature Engineering
- Dashboard Architecture
- Streamlit Dashboard Design
- Power BI Dashboard Design
- Cross-Platform Validation
- Key Insights
- Conclusion

Introduction

Employee engagement directly impacts organizational performance, innovation, and employee retention. Engaged employees demonstrate higher productivity, better collaboration, and lower turnover rates. Conversely, prolonged overtime and poor work-life balance can lead to burnout, resulting in reduced efficiency and increased attrition. Organizations often lack centralized analytical tools to monitor engagement and burnout patterns in real time. Traditional reporting methods are static and do not allow dynamic filtering or department-level insights.

This project addresses the gap by developing interactive dashboards that visualise engagement metrics and burnout classifications, enabling management to identify risks early and implement corrective strategies.

Executive Summary

This project presents a comprehensive analytics solution designed to monitor employee engagement and burnout risk using interactive dashboards developed in Power BI and Streamlit. The objective was to transform raw employee-level satisfaction data into actionable organizational insights.

A composite Engagement Index was engineered from multiple satisfaction metrics, and a rule-based burnout classification model was implemented using overtime and work-life balance indicators. The solution enables HR stakeholders to proactively identify engagement trends, high-risk departments, and burnout-prone employee groups.

The project demonstrates analytical modeling, feature engineering, data visualization, and cross-platform consistency between enterprise BI tools and Python-based web dashboards.

Background & Business Context

Employee engagement is a multidimensional construct influenced by involvement, satisfaction, workplace environment, and interpersonal relationships. High engagement correlates with improved productivity, lower attrition, and enhanced workplace culture.

Burnout, often driven by excessive overtime and poor work-life balance, negatively affects employee morale and organizational performance.

Traditional HR reporting methods rely on static spreadsheets and manual aggregation. These approaches lack:

- Real-time filtering
- Multi-dimensional comparison
- Department-level analysis
- Risk classification mechanisms

This project addresses these limitations through a structured analytical framework and dynamic visualization interface.

Objectives

The primary objectives of this project were:

1. Develop a measurable Engagement Index.
2. Create a burnout risk classification system.
3. Build interactive dashboards for exploratory analysis.
4. Ensure analytical consistency across Power BI and Streamlit.
5. Provide actionable HR insights.

Dataset Overview

The dataset consists of employee-level records containing demographic and workplace satisfaction attributes.

Key Variables Utilized:

Variable	Purpose
JobInvolvement	Measures employee commitment level
JobSatisfaction	Overall job contentment
EnvironmentSatisfaction	Workplace condition satisfaction
RelationshipSatisfaction	Interpersonal satisfaction
WorkLifeBalance	

	Balance between personal and professional life
OverTime	Indicator of extended working hours
Department	Organizational unit classification

Each row represents an individual employee observation.

Data Preparation & Feature Engineering

Data Loading:

In Streamlit, the dataset was loaded using a cached data function to improve performance:

```
@st.cache_data
def load_data():
    df = pd.read_csv("data/Palo_Alto_Networks.csv")
    return df
```

Caching ensures:

- Faster reload time
- Efficient memory utilization
- Reduced redundant computations

Engagement Index Construction:

A composite engagement metric was engineered to unify multiple satisfaction indicators into a single interpretable score.

Formula:

Engagement Index = (JobInvolvement + JobSatisfaction + EnvironmentSatisfaction + RelationshipSatisfaction) / 4

```
df["Engagement_Index"] = (
    df["JobInvolvement"] +
    df["JobSatisfaction"] +
    df["EnvironmentSatisfaction"] +
    df["RelationshipSatisfaction"]
) / 4
```

Implementation in Power BI (DAX):

```
Engagement Index =
(
    'Palo_Alto_Networks'[JobInvolvement] +
    'Palo_Alto_Networks'[JobSatisfaction] +
    'Palo_Alto_Networks'[EnvironmentSatisfaction] +
    'Palo_Alto_Networks'[RelationshipSatisfaction]
) / 4
```

This metric allows simplified interpretation while preserving multidimensional satisfaction influence.

Burnout Classification Logic:

A rule-based classification approach was adopted to ensure interpretability and practical applicability.

Business Logic:

- High Burnout → Overtime = Yes AND WorkLifeBalance ≤ 2
- Medium Burnout → Overtime = Yes
- Low Burnout → Otherwise

Python Implementation:

```
def burnout_level(row):
    if row["OverTime"] == "Yes" and row["WorkLifeBalance"] <= 2:
        return "High"
    elif row["OverTime"] == "Yes":
        return "Medium"
    else:
        return "Low"


df["Burnout_Level"] = df.apply(burnout_level, axis=1)
```

Power BI Implementation (DAX):

```
Burnout Level =
IF(
    'Palo_Alto_Networks'[OverTime] = "Yes"
        && 'Palo_Alto_Networks'[WorkLifeBalance] <= 2,
    "High",
    IF(
        'Palo_Alto_Networks'[OverTime] = "Yes",
        "Medium",
        "Low"
    )
)
```

This deterministic logic ensures transparency and explainability.

Dashboard Architecture:

Two parallel implementations were developed:

- Enterprise BI (Power BI)
- Python-based Web Application (Streamlit)

Streamlit Dashboard Design

Application Structure:

The Streamlit application was modularly structured into:

1. Imports
2. Page configuration

3. Data loading
4. Feature engineering
5. Filtering logic
6. KPI rendering
7. Visualization rendering

This structured approach improves maintainability and readability.

Sidebar Filtering

department_filter = st.sidebar.selectbox(...)

overtime_filter = st.sidebar.selectbox(...)

min_engagement = st.sidebar.slider(...)

Filters dynamically update all KPIs and visualizations using context-aware dataframe slicing.

KPI Metrics

st.metric("Average Engagement Index", ...)

st.metric("Average Work-Life Balance", ...)

st.metric("High Burnout %", ...)

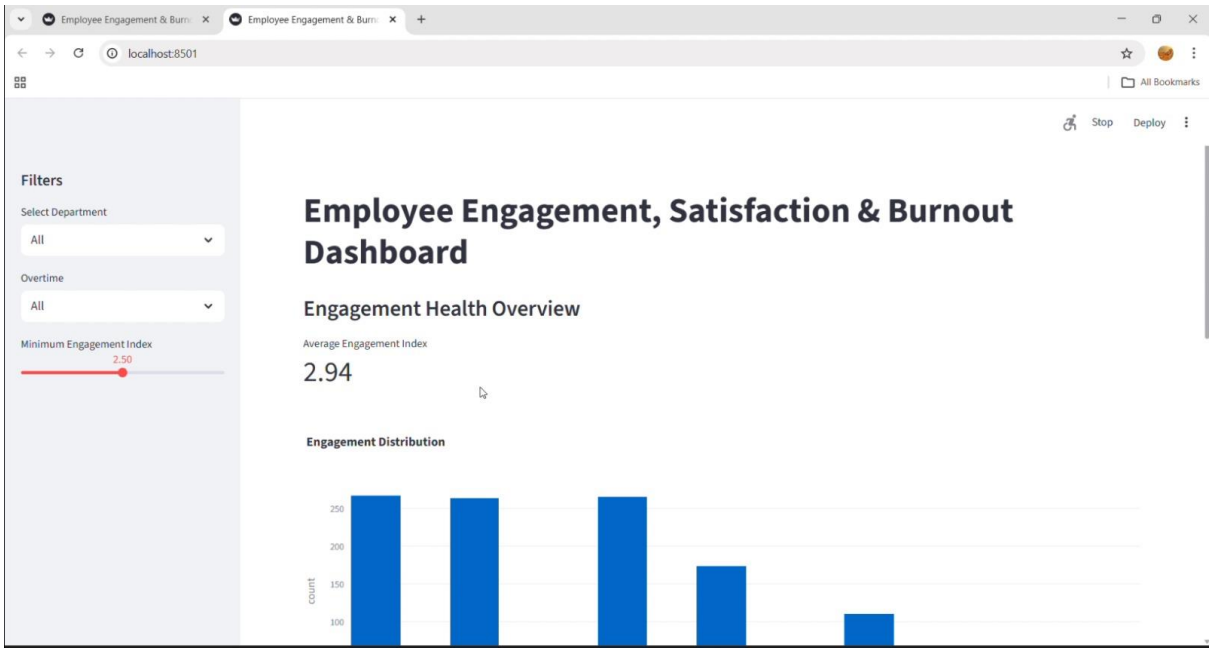
These provide executive-level summary indicators.

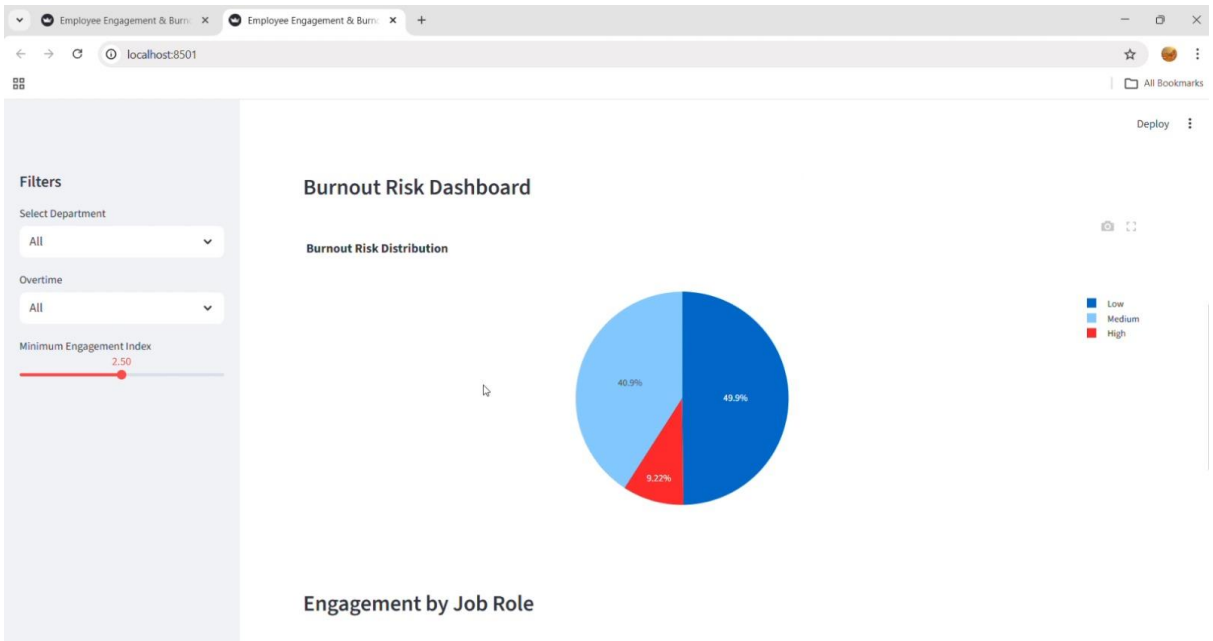
Visualizations

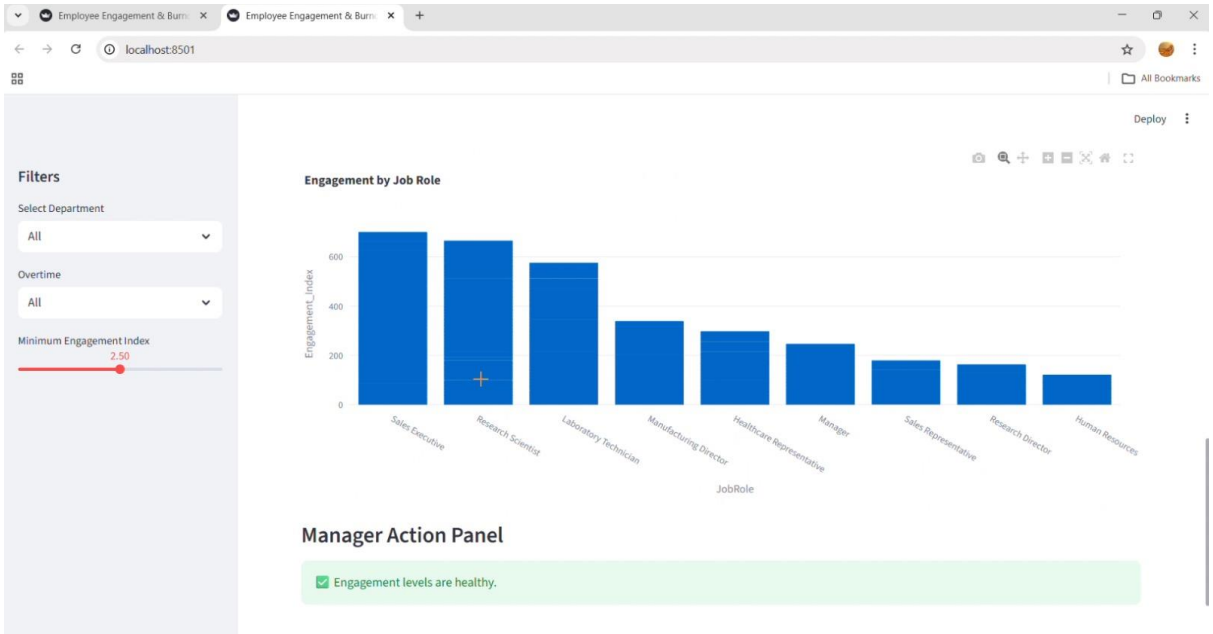
- Histogram → Engagement Distribution
- Bar Chart → Engagement by Department

- Pie Chart → Burnout Distribution
- Expandable Data Table → Transparency

All charts are dynamically responsive to filtering conditions.







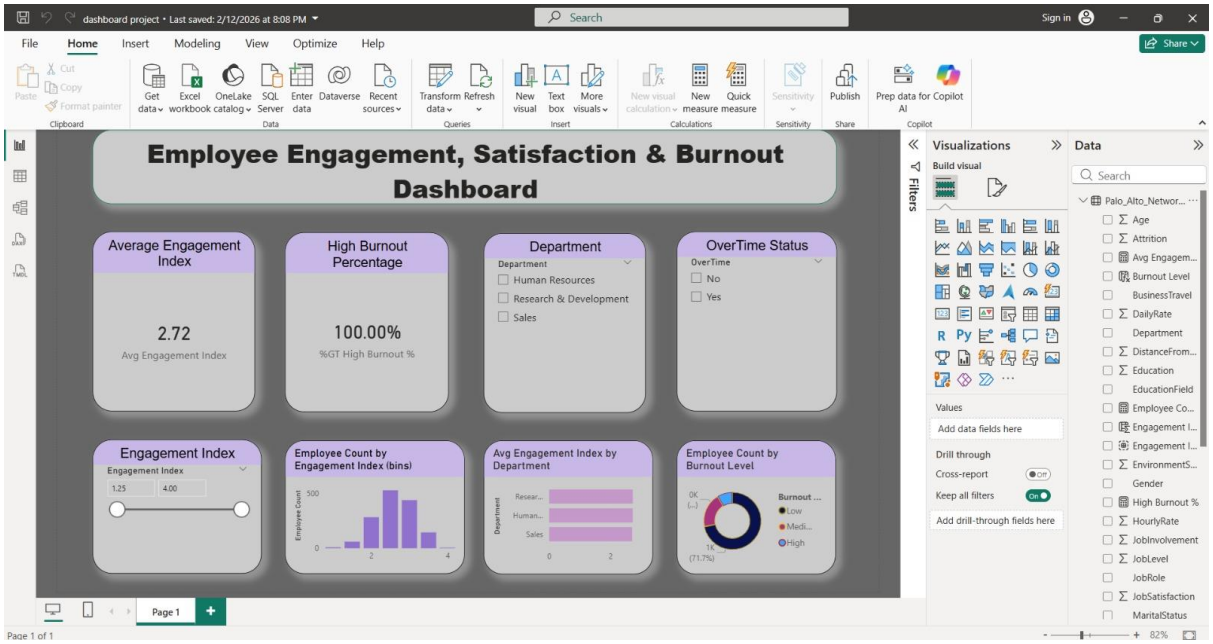
Power BI Dashboard Design

Power BI implementation mirrors the analytical logic of Streamlit.

Visual Components:

- Card visuals → KPI indicators
- Column chart → Engagement Distribution
- Clustered bar chart → Department Comparison
- Donut chart → Burnout Distribution
- Slicers → Interactive filtering

DAX measures ensure contextual aggregation.



Cross-Platform Validation:

Analysis Objective	Power BI	Streamlit
KPI Monitoring	Card	Metric
Filtering	Slicer	Sidebar
Distribution Analysis	Column Chart	Histogram
Department Comparison	Bar Chart	Bar Chart
Burnout Classification	Donut Chart	Pie Chart

This validates analytical consistency across platforms.

Key Insights

- Engagement levels vary significantly across departments.
- Overtime strongly correlates with burnout classification.
- Work-life balance acts as a protective factor.
- High burnout segments can be isolated through dynamic filtering.

Conclusion:

The project successfully demonstrates how raw employee data can be transformed into actionable engagement and burnout analytics using both enterprise BI and Python-based visualization tools. The system provides transparency, interactivity, and cross-platform consistency while maintaining interpretability.

This solution enables proactive HR decision-making and establishes a scalable framework for workforce analytics.